

5(a)	<table border="1"> <thead> <tr> <th>ion</th><th>direction of movement of ion</th><th>action resulting from movement of ion</th></tr> </thead> <tbody> <tr> <td>Na⁺</td><td>from lumen of proximal convoluted tubule into proximal convoluted tubule cell</td><td>co-transport / secondary active transport, of, amino acids / glucose or osmosis of water ;</td></tr> <tr> <td>K⁺ ;</td><td>into guard cell</td><td>opening of stoma(ta)</td></tr> <tr> <td>Na⁺ ;</td><td>into presynaptic knob</td><td>depolarisation of presynaptic membrane</td></tr> <tr> <td>Ca²⁺ ;</td><td>into presynaptic knob</td><td>exocytosis of acetylcholine</td></tr> <tr> <td>H⁺</td><td>from intermembrane space into matrix of mitochondrion</td><td>ATP synthesis ;</td></tr> <tr> <td>Ca²⁺</td><td>from sarcoplasmic reticulum to cytoplasm of muscle fibre</td><td>binds to troponin / troponin shape changes / tropomyosin moves / actin binding site on myosin head exposed / binding of myosin head to actin / allows power stroke / sarcomere contracts ;</td></tr> </tbody> </table>	ion	direction of movement of ion	action resulting from movement of ion	Na ⁺	from lumen of proximal convoluted tubule into proximal convoluted tubule cell	co-transport / secondary active transport, of, amino acids / glucose or osmosis of water ;	K ⁺ ;	into guard cell	opening of stoma(ta)	Na ⁺ ;	into presynaptic knob	depolarisation of presynaptic membrane	Ca ²⁺ ;	into presynaptic knob	exocytosis of acetylcholine	H ⁺	from intermembrane space into matrix of mitochondrion	ATP synthesis ;	Ca ²⁺	from sarcoplasmic reticulum to cytoplasm of muscle fibre	binds to troponin / troponin shape changes / tropomyosin moves / actin binding site on myosin head exposed / binding of myosin head to actin / allows power stroke / sarcomere contracts ;	6
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5(b)(i)	Ca ²⁺ / calcium ions ;	1																					
5(b)(ii)	to avoid wasting <u>energy</u> or <i>idea of</i> avoid closing, due to a single contact / by an object such as water drop ;	1																					
5(b)(iii)	<u>exocytosis</u> ;	1																					
5(b)(iv)	not enough, nitrogen / nitrate, in the soil or poor mineral content of the soil ; R nutrients	1																					